

GALGOTIAS UNIVERSITY

SCHOOL OF ARTIFICIAL INTELLIGENCE

Building Future-Ready AI Talent

LABORATORY ASSIGNMENT

Course Name	Database Management System		
Course Code	E1UC302B	Program	B.Tech. (AIML)
Semester / Batch	III	Academic Year	2026–27
Lab Assignment No.	01	Date	
Faculty In-Charge	Mr. Arunendra Mani Tripathi	Max. Marks	10
Week No.	2	Duration	90 minutes

Course Outcome(s)	CO1: Conceptual understanding of DBMS, ability to define and manipulate data, understanding data independence, and the overall structure of databases CO2: Ability to design and query relational databases using SQL, ensure data integrity, and apply relational algebra and calculus concepts in practical scenarios.
Skill(s) Addressed	SQL Programming, DDL & DML Operations, Database Object Management

1. GENERAL INSTRUCTIONS

- Attempt all questions individually.
- Write and execute SQL queries in MySQL Workbench.
- Save the SQL script as RollNo_Name_DBMS_Lab_SetX.sql.
- Each task must be supported by the executed SQL query and corresponding output.
- Use appropriate SQL data types and meaningful names.
- Submit the SQL script and required screenshots through the LMS.
- Plagiarised or copied submissions will be awarded zero.

2. PROBLEM STATEMENT

A company wants to maintain and update employee records.

3. TASKS / QUESTIONS

- Q1.** Create CompanyDB and an Employee table with Emp_ID, Emp_Name, Department, Salary and Joining_Date using suitable data types.
- Q2.** Insert at least 8 employee records.
- Q3.** Add an Email column using ALTER TABLE.
- Q4.** Increase the salary of IT department employees by 10% using UPDATE.
- Q5.** Delete employees whose salary is below ₹25,000.
- Q6.** Rename Employee as Employee_Details.

Q7. Display the final records.

Q8. Demonstrate TRUNCATE TABLE and DROP TABLE after recording the final result.

4. STUDY MATERIAL / REFERENCE

- MySQL 8.0 Reference Manual: <https://dev.mysql.com/doc/refman/8.0/en/>
- CREATE TABLE: <https://dev.mysql.com/doc/refman/8.0/en/create-table.html>
- ALTER TABLE: <https://dev.mysql.com/doc/refman/8.0/en/alter-table.html>
- Data Manipulation Statements: <https://dev.mysql.com/doc/refman/8.0/en/sql-data-manipulation-statements.html>

5. EVALUATION RUBRIC

Marks are awarded question-wise against the criteria below; partial credit applies for partially correct logic or output.

Evaluation Criteria	Marks
Database and table creation with appropriate data types	2
Correct INSERT and data population	2
ALTER and RENAME operations	1
UPDATE operations	1
DELETE operation	1
TRUNCATE and DROP demonstration	1
Output verification and viva	2
TOTAL	10

6. EVALUATION PROCESS

- Submission check – SQL script and required outputs received within the deadline.
- Execution check – queries are executed successfully and outputs are verified.
- Rubric scoring – marks are awarded against the stated criteria.
- Viva/spot-check – brief oral questions may be asked to verify individual understanding.

Grading Scale

Marks Range	Rating	Descriptor
9 – 10	Excellent	All tasks correct; clean, well-documented code
7 – 8	Good	Minor errors; logic and outputs largely correct
6 – 7	Satisfactory	Partial completion; conceptual gaps evident
Below 5	Needs Improvement	Incomplete or largely incorrect submission